

B1.2 What Happens in Cells

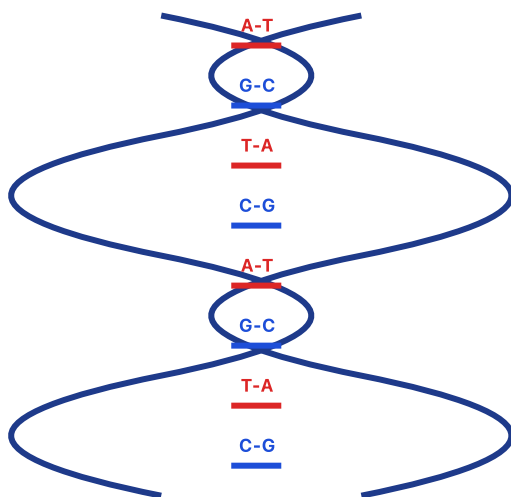
OCR A GCSE Biology (J247) Gateway — Topic B1: Cell-level systems

1. The Structure of DNA

DNA (deoxyribonucleic acid) is the molecule that stores the genetic instructions for life. It is found in the **nucleus** of eukaryotic cells, packaged into structures called **chromosomes**. A short section of DNA that codes for a particular protein is called a **gene**.

DNA is a **double helix** — two strands twisted around each other. Each strand is a chain of repeating units called **nucleotides**, and each nucleotide is made of a sugar (deoxyribose), a phosphate group, and one of four **bases**: A, T, C or G.

DNA Double Helix & Base Pairing



Complementary base pairs

A — T (Adenine ↔ Thymine)

G — C (Guanine ↔ Cytosine)

2 H-bonds between A & T

3 H-bonds between G & C

Backbone:

phosphate + deoxyribose sugar

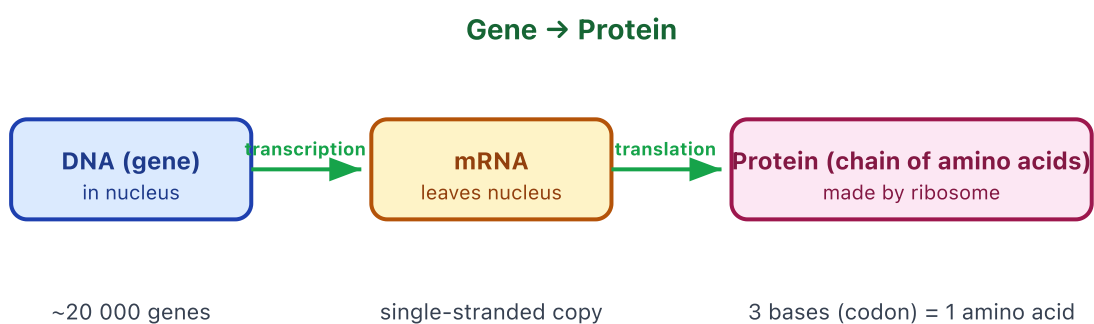
Each "rung" = a base pair

Complementary base pairing: A always pairs with T (held by 2 hydrogen bonds), and G always pairs with C (held by 3 hydrogen bonds). This means knowing the sequence of one strand tells you the sequence of the other.

Exam Tip: Always remember the rule "**A pairs with T, G pairs with C**". A common trick: "Apple in the Tree, Car in the Garage".

From gene to protein — overview

A gene's sequence of bases codes for the order of amino acids in a protein. Three bases (a **triplet** or codon) code for one amino acid. There are 20 different amino acids, joined together to make different proteins. Different sequences of amino acids fold into different 3D shapes — this gives each protein its specific function.



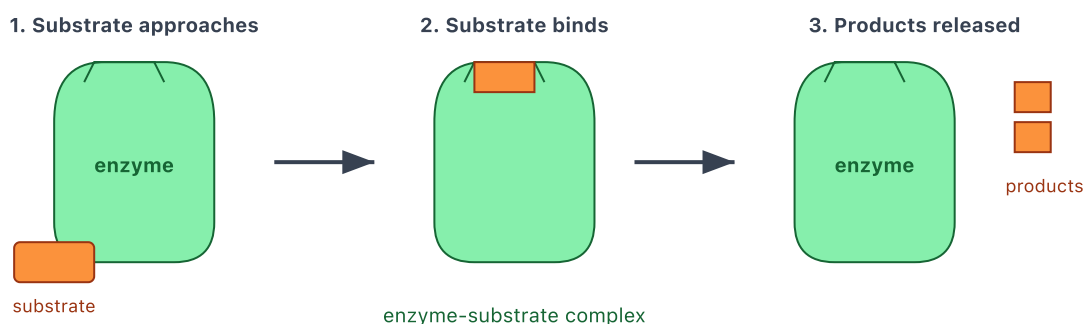
Each protein folds into a specific 3D shape giving it a function.

2. Enzymes

An **enzyme** is a **biological catalyst** — a protein that speeds up a chemical reaction without being used up itself. Enzymes are needed because chemical reactions in cells happen too slowly at body temperature to keep an organism alive.

Each enzyme is a specific shape. The reactant (the **substrate**) fits into a region called the **active site**. This is the **lock-and-key** model: only a substrate of the right shape will fit into the enzyme's active site, just like a key into a lock. After the reaction, the products are released and the enzyme is unchanged, ready to bind another substrate.

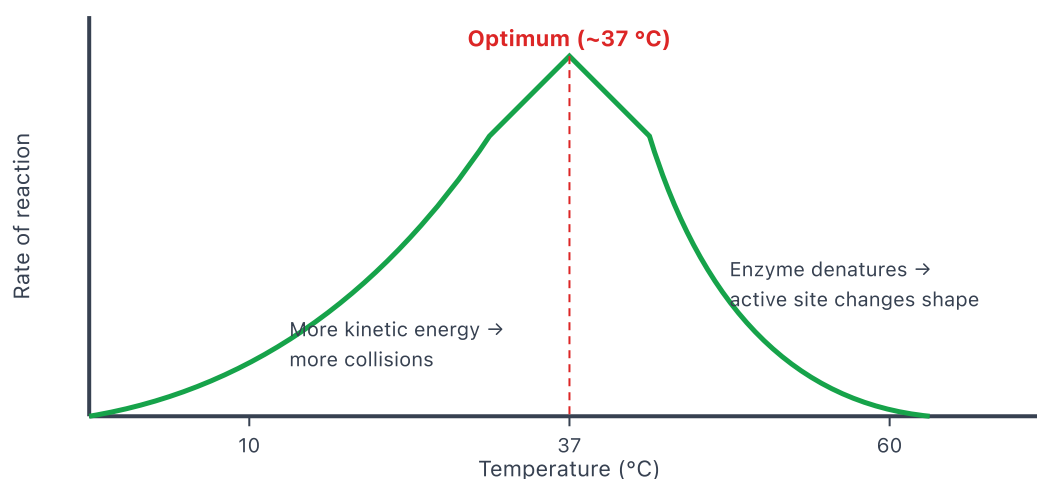
Lock-and-Key Model of Enzyme Action



Factors affecting enzyme activity

Factor	Effect on rate
Temperature	Rate increases as temperature rises (more kinetic energy → more successful collisions). Above the optimum (~37 °C in humans), the enzyme starts to denature : hydrogen bonds break, the active site changes shape and the substrate no longer fits. Rate falls rapidly.
pH	Each enzyme has an optimum pH (e.g. pepsin pH 2 in the stomach; amylase pH 7 in the mouth; lipase pH 8 in the small intestine). Outside this range bonds in the active site break and the enzyme denatures.
Substrate concentration	Rate increases as substrate concentration increases (more collisions). Eventually a plateau is reached — all active sites are occupied; rate is now limited by enzyme concentration.
Enzyme concentration	Rate increases as enzyme concentration increases (more active sites available), provided substrate is in excess.

Effect of Temperature on Enzyme Activity



Exam Tip: When asked to **describe** an enzyme graph: say WHAT happens (rate increases / peaks / drops). When asked to **explain**: relate to molecules — kinetic energy and collisions, denaturing of the active site so substrate no longer fits.

3. Calculating Rate of Reaction

$$\text{Rate} = \frac{\text{amount of substance changed}}{\text{time}}$$

For example, if 30 cm³ of oxygen is produced in 60 seconds, the rate is $30 \div 60 = 0.5 \text{ cm}^3/\text{s}$. Practise these in the practical questions for B1.2.

4. Required Practical Link

Linked PAG: **investigate the effect of pH (or temperature) on amylase activity**. Use iodine test to time how long it takes for amylase to fully break down starch into sugars (when iodine no longer turns blue/black). See PAG notes for full method.